

Which Pair Is It About?

Query-Conditioned Verification in a Deployed Biotic-Interaction Extraction Pipeline

Anonymous ACL submission

Abstract

Literature-mining pipelines that populate biotic-interaction databases propose candidate (taxon, relation, taxon) triples with a supporting passage, then filter them with a sentence-level interaction classifier before curation. We audit that filter on 100 candidates from a deployed system, graded by a domain expert, 65 of which are actually supported by their passage. The filter is close to inert by construction: a rule layer runs first, so every one of the 175,588 candidates in the retrieval pool already carries two taxon surface forms and an interaction surface form. It accepts 91 of 100 and reaches precision 0.714, not significantly above the 0.650 obtained by accepting everything (one-sided exact binomial $p = 0.119$). We decompose its 26 false positives by mechanism: 11 pair-binding failures, in which every component is correct but the relation is not asserted between *this* pair; 8 predicate misbindings; and 7 entity failures. Conditioning the same decision on the candidate triple repairs 22 of the 26: a distilled 110M cross-encoder reading the triple as a query reaches precision 0.934 and F1 0.905 (McNemar $p = 0.018$), and recovers 6 of the 9 leaked true positives among 50 curated pipeline rejections. We state the limit plainly: at the deployed filter’s own acceptance level the verifier reproduces its confusion matrix, so the contribution is a tunable calibrated score where a fixed verdict stood. Entity disambiguation and argument order survive the reformulation; we make no direction claim.

1 Introduction

Biotic-interaction databases such as GloBI (Poe-len et al., 2014) are populated in part by text mining. A retrieval system proposes candidate triples (s_1, r, s_2) together with the passage that allegedly supports them, and a filter decides which candidates are worth a curator’s attention. That filter is almost always a *sentence-level* interaction classifier: it is shown the passage and asked whether it

describes a biotic interaction.

On 100 expert-graded candidates from a deployed system, that filter accepts 91. Its precision, 0.714, is higher than the 0.650 obtained by accepting every candidate, but not significantly so (one-sided exact binomial $p = 0.119$); its recall on supported triples is 1.000. This is not a tuning accident, and it is not carelessness by the people who built it. A rule layer runs upstream of the classifier, and every candidate it emits already contains two taxon surface forms and an interaction surface form — on all 175,588 rows of the retrieval pool. The property the classifier was trained to detect is therefore, as a lexical matter, constant across its entire deployment distribution. It was trained where negatives lack interaction language; at deployment no negative lacks it.

What the pipeline actually needs to know is narrower: not whether the passage describes an interaction, but whether it asserts *this* relation between *these two* taxa. We decompose the filter’s 26 false positives along that distinction and find three mechanisms (§4): 11 pair-binding failures, where every component is individually correct and only the binding is wrong; 8 predicate misbindings, where the interaction term’s real arguments are other tokens; and 7 entity failures. The first class is invisible to any checker that validates s_1, r and s_2 separately, which bounds every component-wise architecture at precision 0.802 at full recall.

Conditioning the decision on the candidate triple repairs 22 of the 26. A 110M cross-encoder, distilled from a teacher that was itself asked the triple-specific question, reaches precision 0.934 at recall 0.877 (§6), clearing the component-wise bound by 0.154 at its own recall, and recovers true positives the pipeline had already discarded. Removing the query from its input at inference collapses its ranking to near the base rate, which is the controlled evidence that the reformulation rather than the encoder is doing the work.

We also state the ceiling, because a reviewer will find it otherwise. At 91 accepts — the deployed filter’s own operating point — the verifier reproduces its confusion matrix. With 65 supported triples among 100, any system accepting 91 incurs at least 26 false positives, and both systems attain that floor. The contribution is therefore not superiority at the deployed operating point, which is unavailable to anyone, but the existence of an operating curve where a fixed verdict stood: a curator can take 84 candidates at precision 0.774 with no recall loss at all, or 61 at 0.934.

Finally, scope: of 50 curated pipeline rejections only 11 are classifier errors, the rest failing upstream at the rule layer or at taxon resolution. A classifier contribution addresses a minority of this pipeline’s rejection volume, and we claim no more.

Contributions.

- An audit of a deployed filter against its own expert-graded output, with the structural reason it accepts almost everything (§2–§3).
- A mechanism decomposition of its false positives, a bound on component-wise remedies, and a measurement of what the reformulation repairs (§4, §6).
- An operating-point analysis reported against our own interest, and a recall-side benchmark with its leakage audited (§6.3, §6.4).
- Four negative results and a measurement of why argument order is not currently evaluable (§7, §8).

2 Setting

The pipeline under audit. BiotXplorer (Ruch et al., 2024) proposes candidate biotic-interaction triples from the literature indexed by SIB Literature Services (Gobeill et al., 2020). For each passage it matches taxon surface forms against a taxonomy and interaction surface forms against a vocabulary derived from the Relation Ontology, and emits every resulting (s_1, r, s_2) combination together with the passage, the three matched surface strings, their canonical forms, and heuristic retrieval scores. Candidates are therefore proposed by *co-occurrence*: a passage naming two taxa near an interaction term produces a candidate whether or not it asserts that interaction between those two. A filter then decides what reaches a curator, and in the deployed system that filter is a sentence-level interaction classifier.

Why the classifier’s feature is inert at this stage.

The rule layer runs before the classifier, and it is also what defines the pool. We pulled the deployment pool in full: 175,588 candidate rows over 165,325 distinct passages and 115,404 distinct triplet keys. Both taxon surface forms and the interaction surface form are non-empty on 175,588 of 175,588 rows, because the presence of an interaction term is the pool’s construction filter; an independent lexical scan for GloBI interaction vocabulary fires on 99.82% of the same passages. Every input the classifier sees at deployment therefore satisfies, lexically, the property it was trained to discriminate. The generalisation is a partial-input failure whose cue was installed by the pipeline’s own architecture rather than by an annotator: the missing input is the query the pipeline itself issued.

Benchmarks. BIOTX100 is 100 retrieved candidates graded by a domain expert on four axes: whether each taxon surface form was correctly resolved (92/100 and 98/100), whether the relation was correct (90/100), and whether the whole triple is supported by its passage (65/100). It uniquely records the deployed filter’s own accept/reject decision, which is what makes §3 possible. Every row has rank 1, so this is precision at one retrieved candidate. REJECT50 is 50 curated pipeline *rejections*, 11 of them true positives wrongly discarded; it is the only recall-side instrument in this paper, and the deployed filter scores 0/11 on it by construction. TEST287 is a larger pair-level set assembled from expert curation: 287 items with a recoverable relation term, 0.544 positive, used here for ranking only.

A query-representation correction. Each candidate carries both a canonical string (`species1_term`) and the surface form matched in the passage (`species1_form`). These differ on 34, 38 and 59 of 100 rows for s_1 , s_2 and r , and the canonical string is *absent from the passage* on 31, 27 and 54 rows. Scoring a verifier with canonical strings asks it about text it cannot see. An earlier version of this work did exactly that; every number reported here uses the surface-form query, the representation the model consumes at inference, and comparisons to our own prior reports are void.

Evaluation protocol. Decision thresholds are fixed before evaluation and never selected on the set being reported, coming from a held-out develop-

System	TP	FP	FN	TN	P	F1
Accept everything	65	35	0	0	0.650	0.788
Deployed filter	65	26	0	9	0.714	0.833
Verifier @ 0.219	57	4	8	31	0.934	0.905

Table 1: BIOTX100, 65 of 100 candidates supported. The deployed filter’s precision is not significantly above accepting everything (one-sided exact binomial $p = 0.119$). The verifier is significantly better than the deployed filter on decision correctness (McNemar with continuity correction, $p = 0.018$; exact binomial on the 22/8 discordant pairs, $p = 0.016$). Verifier figures are means over three seeds.

ment split or a closed-form prior-shift rule. The reported operating point $\tau = 0.219$ is demonstrably not tuned on BIOTX100: the F1-optimal threshold there is ≈ 0.025 , giving F1 0.908 against 0.905. All comparisons report the accept-everything baseline, intervals are 20,000-sample bootstraps or Clopper–Pearson, and model figures are means over three seeds. BIOTX100 shares no passage with training data; REJECT50 shares nine, audited in §6.3.

3 Auditing the deployed filter

Table 1 gives the central observation. The deployed filter accepts 91 of 100 candidates and reaches precision 0.714 against 0.650 for accepting everything, a difference this benchmark cannot resolve ($p = 0.119$; Clopper–Pearson intervals $[0.610, 0.804]$ and $[0.548, 0.743]$). Its recall is 1.000: it rejects nothing that should be kept.

Two things must be said precisely here, because the obvious summary — “the filter does not filter” — is stronger than the data support. First, the filter is not uninformative. Its accept set is not independent of whole-triple support: all nine of its rejections are correct, Fisher’s exact test on its 2×2 decision table gives $p = 3.7 \times 10^{-5}$, and against accepting everything its *accuracy* advantage is significant (exact McNemar on 9 discordant pairs, $p = 0.004$). Second, what it cannot do is convert that signal into precision, because it accepts 91 of 100 candidates and exposes no way to accept fewer.

The sharpest form of the observation conditions on the sentence-level question already being answered affirmatively. Among the 90 candidates whose relation the expert graded correct, the filter accepts 83, and of the 25 that nonetheless fail to support their triple it rejects 7: precision 0.783 against a subset base rate of 0.722, again not sep-

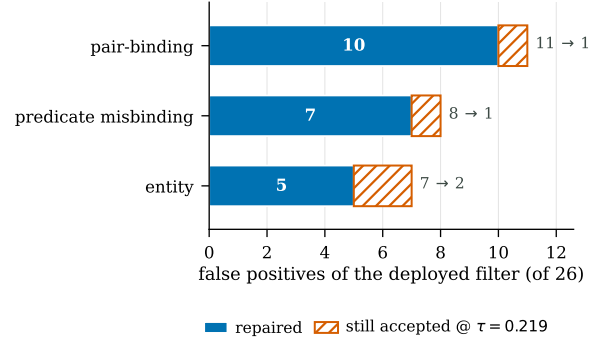


Figure 1: The deployed filter’s 26 false positives by mechanism, and what survives the verifier at $\tau = 0.219$. Twenty-two of 26 are repaired. The three per-class rates (10/11, 7/8, 5/7) are not distinguishable from one another at these cell sizes and we claim no differential.

arable ($p = 0.131$). The verifier on the same 90 reaches 0.950 at 60 kept. Once the sentence-level question is settled, the filter has little discriminative power left over the question the pipeline actually asks.

The signal it does carry is also not the signal it was trained on. Its decisions reach a Matthews correlation of 0.128 with the sentence-level axis it was trained on — does this passage describe a biotic interaction — against 0.560 with whether both taxa were correctly resolved and 0.429 with whole-triple support. A classifier trained to detect interaction language, then deployed where interaction language is a constant (§2), ends up acting on incidental correlates of the question it is not being asked.

4 What the false positives are made of

The deployed filter’s 26 false positives are not homogeneous. We classify each by a fixed cascade over the expert’s three component grades: *entity* if either taxon grade is 0, else *predicate misbinding* if the relation grade is 0, else *pair-binding*. This yields 7 entity, 8 predicate and 11 pair-binding failures; Table 2 gives one worked example of each. Applied to all 35 unsupported candidates rather than to the 26 the filter accepted, the same cascade gives 10, 9 and 16.

The pair-binding class is the one that matters architecturally. Its members have both taxa correctly resolved *and* the correct relation type; the passage simply does not assert that relation between those two. Typically the relation’s true argument is a third organism named in the same sentence, or the sentence lists several taxa of which only a subset

Mechanism (n)	Candidate triple	Passage (abridged)
Pair-binding (11)	(<i>M. marinum</i> , pathogen of, <i>M. tuberculosis</i>)	“ <i>Mycobacterium marinum</i> is a waterborne pathogen of fish and other ectotherms , closely related to <i>Mycobacterium tuberculosis</i> .”
Predicate misbinding (8)	(<i>Salmonella</i> Poona, affected by, cantaloupe)	“Attachment of <i>Salmonella</i> Poona to cantaloupe rind ... as affected by temperature of fruit and inoculum .”
Entity (7)	(<i>P. melanarius</i> , predator, Illiger)	“... the carabid predator <i>Pterostichus melanarius</i> (Illiger) ...”

Table 2: One graded example of each mechanism, from the deployed filter’s 26 false positives. The relation’s true object is italicised where it is not the queried taxon; in the entity case the taxonomic authority’s surname was matched as a taxon.

participate.

Two properties make this class hard for anything short of a joint decision: the relation is genuinely asserted, so relation checking cannot help, and both arguments are genuinely present and correctly resolved, so entity checking cannot either. What is wrong is a binding that exists only in the retrieval step’s combinatorics. The deployed filter is not blind to the class — 5 of its 9 correct rejections are pair-binding failures — it simply rejects too few candidates to matter.

A bound on component-wise verification. This bounds any component-wise architecture at every operating point. A verifier that checks s_1 , r and s_2 separately — however good each check is — rejects at most the 19 component-detectable errors among the 35, and has no signal whatever that separates the 16 pair-binding failures from supported triples. Accepting a fraction r of the 65 supported triples it therefore also accepts all 16, giving precision $65r/(65r + 16)$: 0.802 at full recall and 0.781 at $r = 0.877$. The deployed filter sits 0.088 below that bound at full recall, so it is not merely failing to exploit joint information — it is losing information the components alone would supply. Figure 2 plots the bound.

5 Conditioning the decision on the triple

The reformulation. Putting the pair in the input is not by itself the mechanism; making the *label move when the pair moves* is. In earlier work on this pipeline we added the taxon pair to a sentence-level classifier as a genuine segment pair, changing nothing else, and AUPRC did not move. The reason is that the question the model was *asked* changed while the question its labels *answered* did not: with sentence-level targets, two training items sharing a passage but differing in pair carry the same label, so gradient descent has no reason to attend to the query.

The teacher here is asked, per candidate, whether the passage supports *this* triple, so its labels vary across pairs on the same passage. It is a locally hosted 32B open-weight instruction model (Qwen Team, 2025), prompted zero-shot at temperature 0. On the BIOTX100 items for which we hold human gold it reaches precision 0.919 and F1 0.898, and against 150 human-graded items its overall mislabel rate is $22/150 = 14.7\%$, Clopper–Pearson [9.4%, 21.4%], split 10 false positives and 12 false negatives.

The corpora. VERIFIER-BASE is distilled from 20,000 teacher-labelled rows (0.289 positive). VERIFIER-PC adds three blocks for a total of 48,338 rows (0.343 positive): 16,546 *pair-contrast* rows, 6,201 actively sampled rows, and 5,591 rows in which the candidate’s argument order is reversed. The pair-contrast block is the point of the design: it contributes 2,017 passages that carry *different labels for different pairs*, of which VERIFIER-BASE’s 19,999 distinct passages contain exactly zero. All rows are drawn from the same rule-layer candidate pool as deployment, so 100% carry two taxa and an interaction form and the train/deploy mismatch that produced the original failure is closed by construction.¹

Pair-contrast rows are built by asking the teacher separately about each candidate a pooled passage licenses. Where the labels differ, the passage enters as several rows identical in segment B and differing only in segment A and the target — rows a query-ignoring model cannot fit at all.

Distillation. The student is a 110M-parameter cross-encoder reading the candidate triple as segment A and the passage as segment B, roughly $290\times$ smaller than its teacher, trained with a split

¹These correspond to repository tags V1, V2 and V3 respectively; the pair-contrast block was designed against an earlier, superseded error analysis, so its composition was not targeted at the decomposition reported in §4.

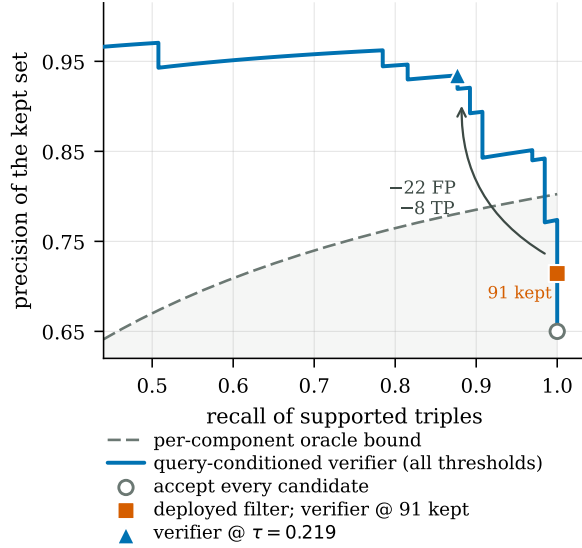


Figure 2: Precision against recall of supported triples on BIOTX100 (three seeds). The dashed curve is the per-component oracle bound $65r/(65r + 16)$ of §4: a verifier with perfect knowledge of both taxa and the relation still admits all 16 pair-binding failures at every operating point. The deployed filter falls below it. The verifier is a curve, not a point, and its 91-candidate operating point coincides with the deployed filter’s confusion matrix.

τ	keep	TP	FP	FN	TN	P	F1	p
0.001	87	65	22	0	13	0.747	0.855	0.387
0.014	80	64	16	1	19	0.800	0.883	0.067
0.025	76	64	12	1	23	0.842	0.908	0.004
0.100	66	59	7	6	28	0.894	0.901	0.016
0.219	61	57	4	8	31	0.934	0.905	0.018
0.850	54	51	3	14	32	0.944	0.857	0.188

Table 3: The verifier’s operating curve on BIOTX100, three seeds. p is McNemar with continuity correction against the deployed filter on decision correctness. The score is usable across a 33-candidate span of acceptance; the $\tau = 0.850$ row is not significant.

Query given to the verifier	P	F1	AUPRC
Retrieved triple (deployed)	0.934	0.905	0.935
Generic relation, pair kept	0.848	0.703	0.852
Relation shuffled across items	0.800	0.114	0.755
Relation only, no taxa	—	—	0.758
Empty query (passage only)	1.000	0.030	0.676

Table 4: Ablating segment A at inference on BIOTX100, weights and threshold unchanged. The base rate is 0.650. With the query removed the same weights rank the benchmark barely above chance. Precision is undefined for the relation-only row, which accepts nothing.

grouped on taxon pair so no pair spans training and development. Three seeds throughout.

6 Results

6.1 Precision side

Table 3 gives the full operating curve. At $\tau = 0.219$ the verifier reaches precision 0.934 and F1 0.905, removing 22 of the deployed filter’s 26 false positives at a cost of 8 true positives, and clearing §4’s per-component bound by 0.154 at its own recall. The precision difference is not a small-sample artefact: Clopper–Pearson intervals $[0.841, 0.982]$ and $[0.610, 0.804]$ do not overlap, the paired bootstrap interval on the difference is $[0.137, 0.308]$ over 20,000 resamples, and perturbing gold labels at random at rates up to 15% leaves the ordering intact in all 20,000 simulations. The F1 difference is far less secure — bootstrap interval $[0.001, 0.145]$ — and we rest no claim on it.

6.2 The query, not the encoder

The deployed filter and the verifier differ in input format, architecture, training corpus and label source at once, so the headline is a deployment comparison and not a controlled ablation. The one factor we *can* isolate is the query, by removing it

at inference with weights and threshold unchanged (Table 4). Ablating segment A collapses ranking from AUPRC 0.935 to 0.676 against a base rate of 0.650, and F1 from 0.905 to 0.030; substituting a generic relation for the retrieved one, pair held fixed, costs 0.202 F1, and shuffling the relation across items costs 0.790. The decision is carried by the query.

This also settles how to read a comparison between our two corpora. VERIFIER-BASE is query-conditioned as well — it differs from VERIFIER-PC in training data, not in input formulation — and on BIOTX100 its global ranking is in fact better: AUPRC 0.966 against 0.935. On TEST287 the ordering reverses (0.905 against 0.921, $P(\Delta > 0) = 0.98$). Pair-contrast supervision buys operating-point behaviour and recall-side recovery, not uniformly better ranking, and we report the cell that does not favour us because omitting it would be selective.

For external reference, the pipeline’s own heuristic retrieval scores used as rankers on BIOTX100 reach AUPRC 0.842, 0.808 and 0.785, and the 32B teacher reaches precision 0.919 and F1 0.898 — which the 110M student matches at 0.934 and 0.905.

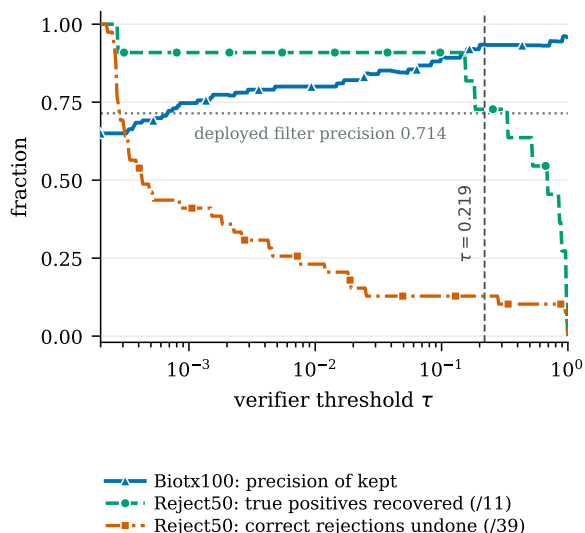


Figure 3: Both sides of the pipeline’s decision against a single threshold. Kept side: precision on BIOTX100. Discarded side: of the 11 true positives among 50 curated pipeline rejections, how many are recovered, and how many of the 39 correct rejections are undone.

6.3 Recall side

REJECT50 asks the complementary question: of the candidates the pipeline threw away, can the verifier find the ones it should have kept? The deployed filter scores 0/11 by construction, having rejected all 50. At $\tau = 0.219$ the verifier recovers 8 of the 11 while wrongly re-admitting 5 of the 39 (Fisher’s exact $p = 2.8 \times 10^{-4}$); at $\tau = 0.10$ it recovers 10/11 at the same cost of 5.

That headline is inflated and we correct it. Nine of the 50 passages also appear in the verifier’s training data, two of them among the 11 true positives, and both leaked positives are recovered. Restricted to the 41 unleaked rows, recovery is 6/9 at $\tau = 0.219$ and 8/9 at $\tau = 0.10$, with 2 of 32 correct rejections undone. The interval on 6/9 is $[0.30, 0.93]$: this demonstrates that recall-side recovery is possible at the operating threshold, and is not a precise rate. BIOTX100 is clean, 0 of 100.

6.4 The operating point, against our own interest

At 91 accepts the verifier’s confusion matrix is 65/26/0/9 — identical to the deployed filter’s. This is arithmetic rather than agreement: with 65 supported triples among 100, any system accepting 91 candidates incurs at least $91 - 65 = 26$ false positives, and both systems attain that floor. The two accept sets are nevertheless not the same, agreeing on 90 of 100 items. We therefore do not

beat the deployed filter at its own operating point, and no system could.

What changes is that its operating point is fixed and ours is not. Three consequences are available to a curator and none were before. At matched *recall* of 1.000 the verifier needs 84 reviews at precision 0.774 rather than 91 at 0.714: seven fewer reviews, no recall cost — though at $n = 100$ this difference is itself not significant ($p = 0.12$). Accepting a 12% recall loss gives 61 reviews at precision 0.934, a third less work at a quarter of the error rate. And a curator who wants everything can still have the 87-candidate point at 0.747. The deliverable is a calibrated score with an adjustable threshold, not a better verdict, and a curator-facing filter should expose the former.

The right threshold is not a property of the model: ingestion into a reference database pays for false positives and belongs near $\tau = 0.85$, while assembling candidates for expert review pays for false negatives and belongs near $\tau = 0.001$, where 87 candidates survive at precision 0.747 and nothing supported is lost. Those points differ by 33 candidates and 0.20 precision. A filter emitting a verdict fixes that choice at training time, for users who were never asked.

7 Where the reformulation stops

Four false positives survive the reformulation: one pair-binding, one predicate, two entity (Figure 1). It is tempting to read the per-class repair rates — 10/11, 7/8, 5/7 — as showing which class the reformulation fixes, and we decline: every pairwise comparison is null (Fisher’s exact $p = 1.00, 0.53, 0.57$), as cell sizes of 11, 8 and 7 make inevitable. The decomposition’s value is the *partition*, not a differential — 16 of the 35 unsupported candidates have every component correct, so no component-wise check reaches them at any operating point — and it is that structural fact the reformulation is designed against.

Entity disambiguation, and a gate that cannot fix it. The natural remedy for the entity class is to gate candidates on taxonomic validity. We tested it against the Open Tree Taxonomy (Rees and Cranston, 2017). It catches 0 of the 7 entity failures and 1 of the 35 unsupported candidates overall, while discarding 5 of the 65 supported ones. The reason is diagnostic rather than incidental: *Illiger*, *Dikarya*, *copia* and *sulla* — the surnames, clades, transposon families and vernacular names

that produced the entity failures — all *resolve* in the taxonomy. The failure is disambiguation, of the kind species-name normalisers are built for (Gerner et al., 2010), not validity. The gate’s casualties are meanwhile viruses (SARS-CoV-2, tomato spotted wilt virus, orchid fleck virus, Colorado tick fever virus), which are legitimate interaction partners in this domain.

Argument order: an open problem, and why.

An ordered query in principle yields direction by argmax over the two orderings. We report no such number, because the gold standard cannot currently score it. Expert curation marks reversed triples as correct — we verified two by hand, (*Breinlia*, host of, *Rattus*) and (*Demidospermus*, parasitized by, *A. inermis*) — and an LLM adjudicator run over adjudicable items contradicts plain surface order in 19 of 64 cases (29.7%, [18.9%, 42.4%]).

The difficulty is systematic rather than random, and this is the part worth reporting. Among the 5,591 teacher-labelled reversals, the rate at which the teacher rejects a reversal is governed by the relation’s *surface string*, not by the relation. Restricted to surface forms with $n \geq 30$, direction-encoding participles and prepositional forms are rejected at 0.51–0.70 (*affecting* 0.70, *affected by* 0.61, *parasite of* 0.57, *feeds on* 0.57), while bare relational nouns are rejected at 0.04–0.10 (*host for* 0.05, *infestation* 0.07, *host of* 0.08, *pollinators* 0.04). Whether the supervision encodes argument order at all depends on which string the rule layer happened to match. A 193-item human direction annotation is in progress; until it exists, a directional score on this data would measure the annotation convention rather than the model. **We make no direction claim in this paper.**

8 Negative results and cost

Table 5 collects four interventions that did not pay, reported because each is a plausible first thing to try. The validity gate is treated in §7. Self-distillation at equal capacity is null, suggesting the teacher’s knowledge is already saturated in the student, which bounds what a same-architecture approach can add. Compression to six layers buys $1.9\times$ CPU throughput for 0.088 precision; since the uncompressed verifier already runs at 24.2 pairs per second on CPU, above this pipeline’s ingestion rate, the trade is available but unattractive. Reweighting the corpus toward the deployment signal distribution cost 0.059 AUPRC on both ar-

Intervention	Benefit	Cost
Taxon-validity gate	0/7 entity failures caught	−5/65 supported triples; rejects viruses
Self-distillation	none	−0.001 P, −0.008 F1, −0.007 AUPRC
6-layer compression	$1.9\times$ (24.2 → 46.2 pairs/s)	−0.088 precision
Corpus signal-matching	none	−0.059 AUPRC, both architectures

Table 5: Four plausible design choices that measurably do not pay.

chitectures we tried.

9 Related work

Filters that answer an easier question. That a classifier can score well by exploiting how its data were assembled is among the best-established results in NLP. Partial-input baselines recover much of the label from the hypothesis alone in natural language inference (Gururangan et al., 2018; Poliak et al., 2018) and from the passage alone in reading comprehension (Kaushik and Lipton, 2018), and such rules hold in distribution while failing under deployment (Geirhos et al., 2020; Du et al., 2024); the closest task-level precedent is fact verification, where a claim-only classifier matches evidence-aware models because of how the corpus was built (Schuster et al., 2019). Section 3 reports that phenomenon rather than a new one. Two things are specific here: the missing input is the query the pipeline itself issued rather than an annotator artefact (Wiegand et al., 2019; Parmar et al., 2023), and the cue the filter does have is inert by construction, since the rule layer fires on every candidate in the pool. A cue present on every input can certify nothing (Feng et al., 2019).

The argument-ignoring failure is already named.

Distant supervision assumes that a sentence containing two related entities expresses the relation (Mintz et al., 2009), and its standard relaxation exists precisely because it does not (Riedel et al., 2010). Since TACRED an instance has been the triple (sentence, e_1 , e_2) rather than the sentence (Zhang et al., 2017), and how the pair is injected into the encoder is worth several F1 points (Baldini Soares et al., 2019; Wu and He, 2019; Zhou and Chen, 2022). Rosenman et al. (2020) name our failure exactly, as an “event heuristic” that classi-

fies on (sentence, relation) while ignoring the arguments, and show that QA-style models repair it on a challenge set. *We claim no part of that diagnosis.* What is ours is that the heuristic is observable in a deployed system’s own expert-graded output rather than in an adversarially constructed set; that the failures partition into classes with a quantified bound on any component-wise remedy (§4); and that we report the operating-point analysis honestly (§6.4). The entity-type shortcut usually invoked to explain such errors (Peng et al., 2020; Mtumbuka and Schockaert, 2024) is unavailable here, because both slots are filled from one inventory of taxon surface forms and type separates nothing.

Attributing failures across pipeline stages is standard, and the accounting is easy to get wrong (Taillé et al., 2020; Ai and Kavuluru, 2023), but it is done across stages on benchmark data; we attribute errors within a production filter’s false positives, against expert grades of the same items, and report against our own interest that only 11 of 50 curated rejections are ours to fix (§6.3).

Conditioning on the query, and what to hand a curator. Casting a relation as a question about a named entity (Levy et al., 2017), or as an entailment hypothesis verbalising the relation between two of them (Obamuyide and Vlachos, 2018; Sainz et al., 2021), is the reformulation our input format instantiates, and the resulting decision — does this passage support this specific claim — has the shape of scientific claim verification (Wadden et al., 2020). Our verifier is a supervised cross-encoder reading the triple as segment A (Nogueira and Cho, 2019), trained on labels from a prompted teacher, as is established practice for turning a large model’s judgments into a small student (West et al., 2022; Gekhman et al., 2023); whether prompting beats supervision at biomedical extraction is contested (Wadhwa et al., 2023; Li et al., 2023; Jimenez Gutierrez et al., 2022; Brokman et al., 2025), but that dispute concerns open-schema label assignment and bears here only on the choice of teacher. The deployment-facing half of the result belongs elsewhere. Selective classification trades coverage against risk by thresholding a score instead of issuing a verdict (Geifman and El-Yaniv, 2017), and that is how automated extraction is made useful to human abstractors: Swaminathan et al. (2024) abstain under calibrated thresholds to automate 57–95% of clinical chart abstraction at controlled error. The filter we audit offers no such

control. What the reformulation buys is not a better point but a curve.

Text mining for interaction databases. GloBI (Poelen et al., 2014) is populated in part by text mining, through semantic-publishing workflows that route mined interactions into the database (Dimitrova et al., 2020); the system audited here (Ruch et al., 2024) retrieves over a biodiversity-indexed MEDLINE (Gobeill et al., 2020). Judging whether an extracted fact is correct before it reaches a curator is an old idea (Rodriguez-Esteban et al., 2006), as is evaluating such tools by whether they help the curator (Hirschman et al., 2012). The closest precedent for our evaluation design is Britan et al. (2018), a deployed curation-support tool scored by the curators of the database it serves against their own annotations; Lee et al. (2018) likewise evaluate a triage model against a resource’s existing practice. Independently of us, Cuzick et al. (2023) work in a framework that takes the interacting pair, rather than the passage, as the annotation unit — the unit our pair-binding class isolates. Large language models are now applied to interaction extraction at scale (Keck et al., 2025; Farrell et al., 2024; D’Souza et al., 2025; Zou et al., 2026), but as extractors: Zou et al. (2026), working over citizen-science comments rather than literature, standardise names and correct categories post hoc without checking extracted interactions back against the text that produced them. We are not aware of published work that verifies literature-mined biotic-interaction triples against their source passage before ingestion, which is the gap this paper occupies.

10 Conclusion

A deployed filter accepts 91 of 100 candidates and reaches a precision this benchmark cannot separate from accepting all of them, because the question it is asked is guaranteed true by the rule layer above it. Its false positives partition into three mechanisms, 16 of the 35 unsupported candidates being invisible to any component-wise check at any operating point. Conditioning the decision on the candidate triple repairs 22 of the 26 and recovers triples the pipeline discarded. At the deployed operating point the two are arithmetically identical: what changes is that a curator has a dial where there was a verdict.

Limitations

Both benchmarks are small and singly annotated. BIOTX100 has 100 graded candidates and REJECT50 has 50; every gold label on both was assigned by one domain expert, no item was graded twice, so we report no inter-annotator agreement and cannot bound label noise. One item moves precision on BIOTX100 by one to two points. The significance of the headline comparison is correspondingly thin: McNemar against the deployed filter gives $p = 0.018$ with continuity correction and $p = 0.016$ exact, and the result keeps its sign under every aggregation we tried, but it would not survive a handful of relabelled items. On the recall side the leaked denominator is 9 positives, and the interval on the reported 6/9 recovery is $[0.30, 0.93]$.

The failure classes are derived, not annotated. Mechanism is assigned by a fixed cascade over the expert’s three component grades rather than by direct annotation, so “pair-binding” is a residual class and any component-grading error accumulates in it. That class is also the one that fixes the 0.802 bound. Cell sizes are 11, 8 and 7, and we claim no differential effectiveness across them (§7).

The deployment comparison is not a controlled ablation. The deployed filter and the verifier differ in input formulation, architecture, training corpus and label source at once. We can show that the query is necessary *within* the verifier (Table 4) and that the filter’s own feature is inert on the deployment pool, but we cannot decompose the $26 \rightarrow 4$ reduction across those four factors. The missing cell is a sentence-only student trained on the same teacher labels, which we did not have the budget to train.

The training labels are wrong about one time in seven. Our training data are labelled by a prompted teacher. Against 150 human-gold items its mislabel rate is 14.7%, $[9.4\%, 21.4\%]$, split 10 false positives and 12 false negatives, and the student inherits that boundary including its systematic errors; training encoders on model-generated labels can degrade and destabilise them (Lu and Smith, 2025). The benchmark is graded by a domain expert rather than by the teacher, so the reported precision is measured after whatever damage the noise does — but the verifier’s ceiling is the teacher’s judgment rather than a curator’s.

Leakage on the recall-side benchmark. Nine of REJECT50’s 50 passages appear in the verifier’s training data, including 2 of its 11 true positives. We report the leakage-free figures in §6.3 and recommend the leaked subset for any future comparison. BIOTX100 is unaffected (0/100).

The classifier owns a minority of the pipeline’s failures. Of 50 curated rejections, 11 are classifier errors. The rest fail upstream: 32 carry an incorrect interaction term and 17 a mis-resolved taxon, 14 of them both, and a further 4 are correct rejections with every component individually right. No classifier change, ours included, addresses the upstream mass, and our precision gain is not a pipeline-level gain. An earlier internal study using a different assignment convention reported a 40/34/22 split across the rule, entity and classifier layers; the overlap-explicit counts above are what we can reproduce, and only the 22% classifier share is stable across conventions.

Argument order is unevaluable on the current gold standard. Expert curation scores reversed triples as correct, so no directional number in this paper would measure the model rather than the annotation convention (§7).

Prior numbers from this project are void. Earlier results were computed with canonical taxon and relation strings that are absent from the passage on up to 54 of 100 rows. Every number here uses the surface-form query; comparisons to our own prior reports should not be made.

One system, one domain, one pretrained teacher. We audit one retrieval system, one interaction vocabulary and one domain, taxon–taxon interactions in MEDLINE, so we cannot show that filters over other pools fail the same way, although the mechanism — a pair-agnostic decision over passages naming more than one candidate pair — is not specific to ours. The passages are published MEDLINE text and the teacher is a pretrained model; we cannot exclude that it saw them, or the interactions they report, during pretraining, so its judgments should be read as an optimistic estimate of performance on genuinely new literature.

References

Xuguang Ai and Ramakanth Kavuluru. 2023. End-to-end models for chemical–protein interaction extraction: Better tokenization and span-based pipeline

strategies. In *2023 IEEE 11th International Conference on Healthcare Informatics (ICHI)*, pages 610–618. IEEE. ArXiv:2304.01344.

Livio Baldini Soares, Nicholas FitzGerald, Jeffrey Ling, and Tom Kwiatkowski. 2019. [Matching the blanks: Distributional similarity for relation learning](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2895–2905, Florence, Italy. Association for Computational Linguistics.

Aurore Britan, Isabelle Cusin, Valérie Hinard, Luc Mottin, Emilie Pasche, Julien Gobeill, Valentine Rech de Laval, Anne Gleizes, Daniel Teixeira, Pierre-André Michel, Patrick Ruch, and Pascale Gaudet. 2018. [Accelerating annotation of articles via automated approaches: evaluation of the neXtA5 curation-support tool by neXtProt](#). *Database*, 2018: bay129.

Aviv Brokman, Xuguang Ai, Yuhang Jiang, Shashank Gupta, and Ramakanth Kavuluru. 2025. [A benchmark for end-to-end zero-shot biomedical relation extraction with LLMs: experiments with OpenAI models](#). In *Proceedings of the Third Workshop for Artificial Intelligence for Scientific Publications*, pages 44–55, Mumbai, India and virtual. Association for Computational Linguistics.

Alayne Cuzick, James Seager, Valerie Wood, Martin Urban, Kim Rutherford, and Kim E. Hammond-Kosack. 2023. [A framework for community curation of inter-species interactions literature](#). *eLife*, 12:e84658.

Mariya Dimitrova, Jorrit Poelen, Georgi Zhelezov, Teodor Georgiev, Donat Agosti, and Lyubomir Penev. 2020. [Semantic publishing enables text mining of biotic interactions](#). *Biodiversity Information Science and Standards*, 4:e59036.

Jennifer D’Souza, Zachary Laubach, Tarek Al Mustafa, Sina Zarrieß, Robert Frühstückl, and Phyllis Illari. 2025. [Mining for species, locations, habitats, and ecosystems from scientific papers in invasion biology: A large-scale exploratory study with large language models](#). In *Proceedings of the 1st Workshop on Ecology, Environment, and Natural Language Processing (NLP4Ecology2025)*, pages 16–23, Tallinn, Estonia. University of Tartu Library.

Mengnan Du, Fengxiang He, Na Zou, Dacheng Tao, and Xia Hu. 2024. [Shortcut learning of large language models in natural language understanding](#). *Communications of the ACM*, 67(1):110–120.

Maxwell J. Farrell, Nicolas Le Guillarme, Liam Brierley, Bronwen Hunter, Daan Scheepens, Anna Willoughby, Andrew Yates, and Nicole Mideo. 2024. [The changing landscape of text mining: a review of approaches for ecology and evolution](#). *Proceedings of the Royal Society B: Biological Sciences*, 291(2027):20240423.

Shi Feng, Eric Wallace, and Jordan Boyd-Graber. 2019. [Misleading failures of partial-input baselines](#). In

Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, pages 5533–5538, Florence, Italy. Association for Computational Linguistics.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, pages 4878–4887.

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. [Shortcut learning in deep neural networks](#). *Nature Machine Intelligence*, 2(11):665–673.

Zorik Gekhman, Jonathan Herzig, Roei Aharoni, Chen Elkind, and Idan Szpektor. 2023. [TrueTeacher: Learning factual consistency evaluation with large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2053–2070, Singapore. Association for Computational Linguistics.

Martin Gerner, Goran Nenadic, and Casey M. Bergman. 2010. [LINNAEUS: A species name identification system for biomedical literature](#). *BMC Bioinformatics*, 11:85.

Julien Gobeill, Déborah Caucheteur, Pierre-André Michel, Luc Mottin, Emilie Pasche, and Patrick Ruch. 2020. [SIB literature services: RESTful customizable search engines in biomedical literature, enriched with automatically mapped biomedical concepts](#). *Nucleic Acids Research*, 48(W1):W12–W16.

Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. 2018. [Annotation artifacts in natural language inference data](#). In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, pages 107–112, New Orleans, Louisiana. Association for Computational Linguistics.

Lynette Hirschman, Gully A. P. C. Burns, Martin Krallinger, Cecilia Arighi, K. Bretonnel Cohen, Alfonso Valencia, Cathy H. Wu, Andrew Chatr-Aryamontri, Karen G. Dowell, Eva Huala, Anália Lourenço, Robert Nash, Anne-Lise Veuthey, Thomas Wieggers, and Andrew G. Winter. 2012. [Text mining for the biocuration workflow](#). *Database*, 2012: bas020.

Bernal Jimenez Gutierrez, Nikolas McNeal, Clayton Washington, You Chen, Lang Li, Huan Sun, and Yu Su. 2022. [Thinking about GPT-3 in-context learning for biomedical IE? think again](#). In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 4497–4512, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Divyansh Kaushik and Zachary C. Lipton. 2018. [How much reading does reading comprehension require? A critical investigation of popular benchmarks](#). In

Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, pages 5010–5015, Brussels, Belgium. Association for Computational Linguistics.

François Keck, Henry Broadbent, and Florian Altermatt. 2025. [Extracting massive ecological data on state and interactions of species using large language models](#). *bioRxiv*. Preprint, posted 27 January 2025.

Kyubum Lee, Maria Livia Famiglietti, Aoife McMahon, Chih-Hsuan Wei, Jacqueline Ann Langdon MacArthur, Sylvain Poux, Lionel Breuza, Alan Bridge, Fiona Cunningham, Ioannis Xenarios, and Zhiyong Lu. 2018. [Scaling up data curation using deep learning: An application to literature triage in genomic variation resources](#). *PLOS Computational Biology*, 14(8):e1006390.

Omer Levy, Minjoon Seo, Eunsol Choi, and Luke Zettlemoyer. 2017. [Zero-shot relation extraction via reading comprehension](#). In *Proceedings of the 21st Conference on Computational Natural Language Learning (CoNLL 2017)*, pages 333–342, Vancouver, Canada. Association for Computational Linguistics.

Guozheng Li, Peng Wang, and Wenjun Ke. 2023. Revisiting large language models as zero-shot relation extractors. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 6877–6892, Singapore. Association for Computational Linguistics.

Yucheng Lu and Kazimier Smith. 2025. Feeding LLM annotations to BERT classifiers at your own risk. *arXiv preprint arXiv:2504.15432*.

Mike Mintz, Steven Bills, Rion Snow, and Dan Jurafsky. 2009. Distant supervision for relation extraction without labeled data. In *Proceedings of ACL-IJCNLP*, pages 1003–1011.

Frank Martin Mtumbuka and Steven Schockaert. 2024. [Entity or relation embeddings? an analysis of encoding strategies for relation extraction](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 6003–6022, Miami, Florida, USA. Association for Computational Linguistics.

Rodrigo Nogueira and Kyunghyun Cho. 2019. [Passage re-ranking with BERT](#). *arXiv preprint arXiv:1901.04085*.

Abiola Obamuyide and Andreas Vlachos. 2018. [Zero-shot relation classification as textual entailment](#). In *Proceedings of the First Workshop on Fact Extraction and VERification (FEVER)*, pages 72–78, Brussels, Belgium. Association for Computational Linguistics.

Mihir Parmar, Swaroop Mishra, Mor Geva, and Chitta Baral. 2023. [Don’t blame the annotator: Bias already starts in the annotation instructions](#). In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 1779–1789, Dubrovnik, Croatia. Association for Computational Linguistics.

Hao Peng, Tianyu Gao, Xu Han, Yankai Lin, Peng Li, Zhiyuan Liu, Maosong Sun, and Jie Zhou. 2020. [Learning from context or names? an empirical study on neural relation extraction](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3661–3672, Online. Association for Computational Linguistics.

Jorrit H Poelen, James D Simons, and Chris J Mungall. 2014. Global biotic interactions: An open infrastructure to share and analyze species-interaction datasets. *Ecological Informatics*, 24:148–159.

Adam Poliak, Jason Naradowsky, Aparajita Haldar, Rachel Rudinger, and Benjamin Van Durme. 2018. [Hypothesis only baselines in natural language inference](#). In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics*, pages 180–191, New Orleans, Louisiana. Association for Computational Linguistics.

Qwen Team. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Jonathan A. Rees and Karen Cranston. 2017. [Automated assembly of a reference taxonomy for phylogenetic data synthesis](#). *Biodiversity Data Journal*, 5:e12581.

Sebastian Riedel, Limin Yao, and Andrew McCallum. 2010. [Modeling relations and their mentions without labeled text](#). In *Machine Learning and Knowledge Discovery in Databases (ECML PKDD 2010)*, Lecture Notes in Computer Science, pages 148–163, Barcelona, Spain. Springer Berlin Heidelberg.

Raul Rodriguez-Esteban, Ivan Iossifov, and Andrey Rzhetsky. 2006. [Imitating manual curation of text-mined facts in biomedicine](#). *PLoS Computational Biology*, 2(9):e118.

Shachar Rosenman, Alon Jacovi, and Yoav Goldberg. 2020. [Exposing shallow heuristics of relation extraction models with challenge data](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3702–3710, Online. Association for Computational Linguistics.

Patrick Ruch, Emilie Pasche, Pierre-André Michel, and Déborah Caucheteur. 2024. [BiotXplorer: Navigating evidence-based biotic interactions](#). *Biodiversity Information Science and Standards*, 8:e135453.

Oscar Sainz, Oier Lopez de Lacalle, Gorka Labaka, Ander Barrena, and Eneko Agirre. 2021. [Label verbalization and entailment for effective zero and few-shot relation extraction](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 1199–1212, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Tal Schuster, Darsh Shah, Yun Jie Serene Yeo, Daniel Roberto Filizzola Ortiz, Enrico Santus, and Regina Barzilay. 2019. [Towards debiasing fact verification](#)

964	models. In <i>Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)</i> , pages 3419–3425, Hong Kong, China. Association for Computational Linguistics.	
965		
966		
967		
968		
969	Akshay Swaminathan, Ivan Lopez, William Wang, Ujwal Srivastava, Edward Tran, Aarohi Bhargava-Shah, Janet Y. Wu, Alexander L. Ren, Kaitlin Caoili, Brandon Bui, Layth Alkhani, Susan Lee, Nathan Mohit, Noel Seo, Nicholas Macedo, Winson Cheng, Charles Liu, Reena Thomas, Jonathan H. Chen, and Olivier Gevaert. 2024. Selective prediction for extracting unstructured clinical data . <i>Journal of the American Medical Informatics Association</i> , 31(1):188–197.	
970		
971		
972		
973		
974		
975		
976		
977		
978		
979	Bruno Taillé, Vincent Guigue, Geoffrey Scuttheeten, and Patrick Gallinari. 2020. Let’s stop incorrect comparisons in end-to-end relation extraction! In <i>Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)</i> , pages 3689–3701. Association for Computational Linguistics.	
980		
981		
982		
983		
984		
985		
986	David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. 2020. Fact or fiction: Verifying scientific claims . In <i>Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)</i> , pages 7534–7550, Online. Association for Computational Linguistics.	
987		
988		
989		
990		
991		
992		
993	Somin Wadhwa, Silvio Amir, and Byron Wallace. 2023. Revisiting relation extraction in the era of large language models . In <i>Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)</i> , pages 15566–15589, Toronto, Canada. Association for Computational Linguistics.	
994		
995		
996		
997		
998		
999		
1000	Peter West, Chandra Bhagavatula, Jack Hessel, Jena D. Hwang, Liwei Jiang, Ronan Le Bras, Ximing Lu, Sean Welleck, and Yejin Choi. 2022. Symbolic knowledge distillation: from general language models to commonsense models . In <i>Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies</i> , pages 4602–4625, Seattle, United States. Association for Computational Linguistics.	
1001		
1002		
1003		
1004		
1005		
1006		
1007		
1008		
1009		
1010	Michael Wiegand, Josef Ruppenhofer, and Thomas Kleinbauer. 2019. Detection of abusive language: the problem of biased datasets . In <i>Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)</i> , pages 602–608, Minneapolis, Minnesota. Association for Computational Linguistics.	
1011		
1012		
1013		
1014		
1015		
1016		
1017		
1018	Shanchan Wu and Yifan He. 2019. Enriching pre-trained language model with entity information for relation classification . In <i>Proceedings of the 28th</i>	
1019		
1020		
	<i>ACM International Conference on Information and Knowledge Management (CIKM)</i> , pages 2361–2364, Beijing, China. Association for Computing Machinery.	1022
		1023
		1024
	Yuhao Zhang, Victor Zhong, Danqi Chen, Gabor Angeli, and Christopher D. Manning. 2017. Position-aware attention and supervised data improve slot filling . In <i>Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing</i> , pages 35–45, Copenhagen, Denmark. Association for Computational Linguistics.	1025
		1026
		1027
		1028
		1029
		1030
		1031
	Wenxuan Zhou and Muhao Chen. 2022. An improved baseline for sentence-level relation extraction . In <i>Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)</i> , pages 161–168, Online only. Association for Computational Linguistics.	1032
		1033
		1034
		1035
		1036
		1037
		1038
		1039
	Heng-Xing Zou, Xiaohao Yang, Thabassum H. Hajamaideen, Olivia J. Stein, Roxanne S. Beltran, Benjamin G. Freeman, Mark Lindquist, Eliot T. Miller, Summer Mengarelli, Charlotte M. Probst, Fernanda S. Valdovinos, Derek Van Berkel, Phoebe L. Zarnetske, Brian C. Weeks, and Kai Zhu. 2026. Large language models unlock the ecology of species interactions . <i>bioRxiv</i> . Preprint, posted 10 February 2026.	1040
		1041
		1042
		1043
		1044
		1045
		1046
		1047